LIMIT & CONTINUITY

PROBLEM :

Find the limit of the following functions:

1. $\lim\limits_{x\to 0}\dfrac{x}{\sqrt{x+1}-1}$

2. $\lim\limits_{x\to 2}\dfrac{2x^2-5x+2}{5x^2-7x-6}$

3. $\lim\limits_{x\to 0}\dfrac{x}{|x|}$

4. $f(x)=\begin{cases}2-x & ,x<1\\ x^2+1 & ,x>1\end{cases}$ find $\lim\limits_{x\to 1} f(x)$

5. $f(x)=\begin{cases}3x-1 & ,x<1\\ 3-x & ,x>1\end{cases}$ find $\lim\limits_{x\to 1} f(x)$

6. $f(x)=\begin{cases}\dfrac{1}{x+2} & ,x<-2\\ x^2-5 & ,-2<x<3\\ \sqrt{x+13} & ,x>3\end{cases}$ find $\lim\limits_{x\to -2} f(x)$ and $\lim\limits_{x\to 3} f(x)$

7. $\lim\limits_{x\to\infty}\dfrac{3x+5}{6x-8}$

8. $\lim\limits_{x\to\infty}\sqrt[3]{\dfrac{3x+5}{6x-8}}$

9. $\lim\limits_{x\to-\infty}\dfrac{4x^2-x}{2x^3-5}$

10. $\lim\limits_{x\to\infty}\left(\sqrt{x^6+5}-x^3\right)$

11. $f(x)=\begin{cases}x^2+1 & ,x>0\\ 1 & ,x=0\\ 1+x & ,x<0\end{cases}$ find $\lim\limits_{x\to 0} f(x)$

12. $\lim\limits_{x\to\infty}\left(\sqrt{x^6+5x^3}-x^3\right)$

13. $f(x)=\begin{cases}e^{\frac{-|x|}{2}} & ,-1<x<0\\ x^2 & ,0<x<2\end{cases}$

    Find $\lim\limits_{x\to 0} f(x)$

14. $f(x)=\begin{cases}x^2 & ,x<1\\ 2.4 & ,x=1\\ x^2+1 & ,x>1\end{cases}$

    Does $\lim\limits_{x\to 1} f(x)$ exist ?

15. $f(x)=\begin{cases}2x+1 & ,x<1\\ 3-x & ,x>1\end{cases}$

    find $\lim\limits_{x\to 1} f(x)$

16. Prove that $\lim\limits_{x\to\infty}\left(1+\dfrac{1}{x}\right)^x=e$

Test the continuity of the following functions:

1. $f(x)=\begin{cases} Cosx & ,x\geq 0 \\ -Cosx & ,x<0 \end{cases}$ at x = 0

2. $f(x)=\begin{cases} xCos(1/x) & ,x\neq 0 \\ 0 & ,x=0 \end{cases}$ at x = 0

3. $f(x)=\begin{cases} e^{1/x} & ,x\neq 0 \\ 1 & ,x=0 \end{cases}$ at x = 0

4. $f(x)=\begin{cases} \sqrt{|x|} & ,x\geq 0 \\ -\sqrt{|x|} & ,x<0 \end{cases}$ at x = 0

5. $f(x)=\begin{cases} e^{\frac{-|x|}{2}} & ,-1<x<0 \\ x^2 & ,0\leq x<2 \end{cases}$ at x = 0

6. $f(X)=\begin{cases} (x-a)Sin\dfrac{1}{x-a} & x\neq a \\ 0 & x=a \end{cases}$ at x = a

7. $f(x)=\begin{cases} xSin(1/x) & ,x\neq 0 \\ 0 & x=0 \end{cases}$ at x = 0

8. $f(x)=\begin{cases} 1 & ,x<0 \\ 1+Sinx & ,0\leq x<\pi/2 \\ 2+(x-\pi/2)^2 & ,x\geq \pi/2 \end{cases}$ at x = 0 and $x=\pi/2$

9. $f(x)=|x|+|x-1|$ at x = 0 and x = 1

10. $f(x)=\begin{cases} \dfrac{|x-3|}{x-3} & ,x\neq 3 \\ 0 & ,x=3 \end{cases}$ at x = 3

11. $f(x)=\begin{cases} (1+x)^{1/x} & ,x\neq 0 \\ 1 & ,x=0 \end{cases}$ at x = 0

12. $f(x)=\begin{cases} \dfrac{e^{1/x^2}}{e^{1/x^2}-1} & ,x\neq 0 \\ 1 & ,x=0 \end{cases}$ at x = 0

Answers

Limit : ① 2

② $\frac{3}{13}$

③ DNE

④ DNE

⑤ 2

⑥ for -2 : LHL $= -\infty$
RHL $= -1$ $\therefore$ DNE

for 3 : 4.

⑦ $\frac{1}{2}$

⑧ $\frac{1}{\sqrt[3]{2}}$

⑨ 0

⑩ 0

⑪ 1

⑫ $\frac{5}{2}$

⑬, ⑭, ⑮ $\Rightarrow$ DNE.

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Continuity:

1. not continuous.

2. continuous.

3. LHL = 0, RHL = $+\infty$. Not continuous.

4. continuous.

5. Not continuous.

6. continuous.

7. continuous.

8. for 0: continuous. for $\frac{\pi}{2}$: continuous.

   (Assume x is in degree)

9. for 0 and 1 continuous.

10. Not continuous.

11. Remember the formula $\lim_{x \to 0} (1+x)^{\frac{1}{x}} = e$.

    Not continuous.

12. Divide by $e^{\frac{1}{x^2}}$. Continuous.

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